



Topology Optimization for Inspectability

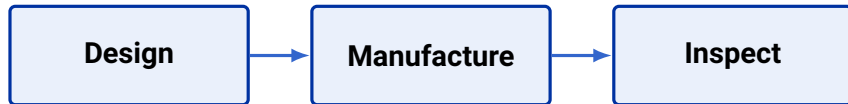
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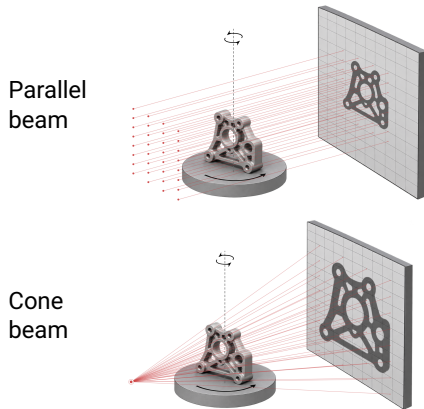
Design for inspection



Manufacturable does not mean inspectable.

Design for X-ray Non-Destructive Evaluation (NDE).

Transmission sets a thickness limit



Beer-Lambert attenuation

$$I = I_0 e^{-\mu x}$$

I_0 : initial intensity I : intensity at the detector
 μ : material coefficient x : material thickness

Require a minimum X-ray intensity
at the detector.

Limit material thickness on each path.

Minimize compliance, constrain inspectability

$$\min_{\rho} C(\rho)$$

Compliance: $\mathbf{f}^T \mathbf{u}$

$$\text{s.t. } K(\tilde{\rho}) \mathbf{u} = \mathbf{f}$$

Elastic equilibrium

$$g_V(\tilde{\rho}) \leq 0$$

Volume

$$g_{A,k}(\tilde{\rho}) \leq 0, \quad k = 1, \dots, N_A$$

Attenuation

$$0 \leq \rho \leq 1$$

Density bounds

ρ : design density $\tilde{\rho}$: filtered density K : stiffness matrix $\mathbf{u}$: displacement $\mathbf{f}$: load

Compute thickness along rays or through a field

Ray tracing

$$a(\mathbf{r}) = \int_0^L \tilde{\rho}(\mathbf{x}(s)) ds$$

MFEM's FindPointsGSLIB

Locate points and interpolate density.

Transport (advection) model

$$\mathbf{v} \cdot \nabla \rho_a = \tilde{\rho} \quad \text{in } \Omega$$

$$\rho_a = 0 \text{ at inflow } \Gamma_-$$

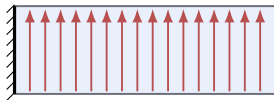
DG with standard MFEM operators.

Read thickness at outflow.

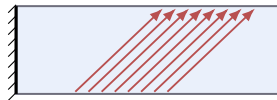
Compare ray tracing and advection



Cantilever load



Vertical rays



Diagonal rays

768 × 256 elements

1,576,962 DOFs

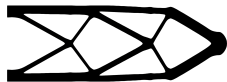
$V_f = 0.5$

Compare ray tracing and advection

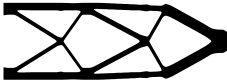
No constraint



Ray tracing



Advection

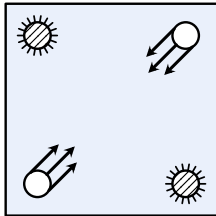


$\alpha_{\max} = 0.5$ vertical

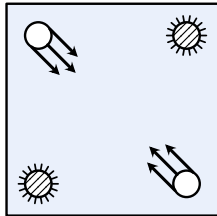
$\alpha_{\max} = 0.3$ vertical

$\alpha_{\max} = 0.3$ diagonal

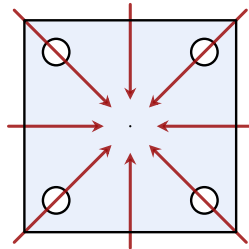
Ray geometry changes the design



Load case 1



Load case 2



Ray directions

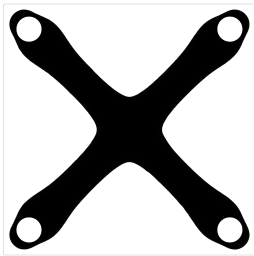
140,800 elements

1,129,978 DOFs

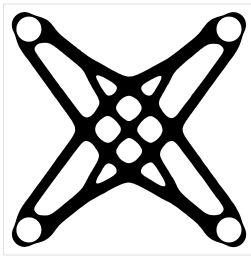
$V_f = 0.35$

$\alpha_{\max} = 0.6$

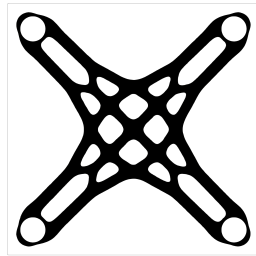
Ray geometry changes the design



No constraint



Parallel beam



Fan beam

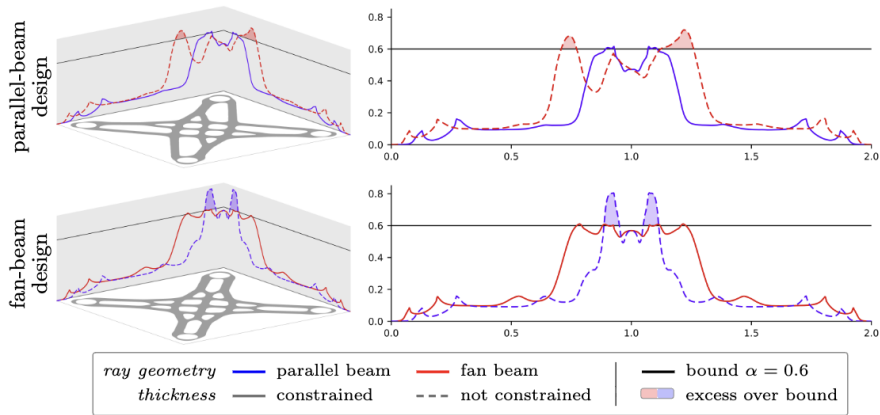
140,800 elements

1,129,978 DOFs

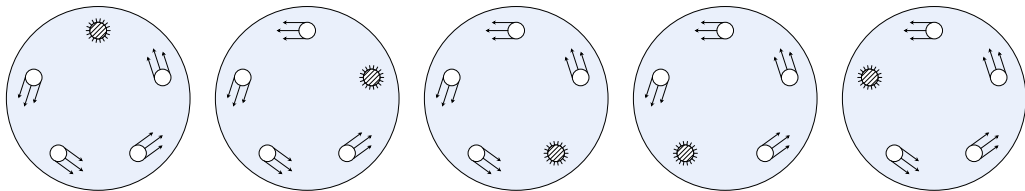
$V_f = 0.35$

$\alpha_{\max} = 0.6$

Ray geometry changes the design



Vary the directions and thickness bound

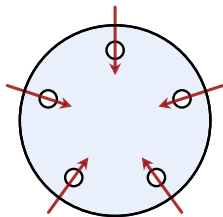


518,144 elements

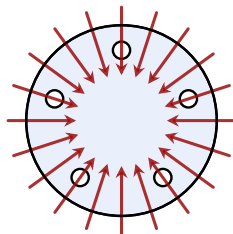
4,663,296 DOFs

$V_f = 0.4$

Vary the directions and thickness bound



5 directions



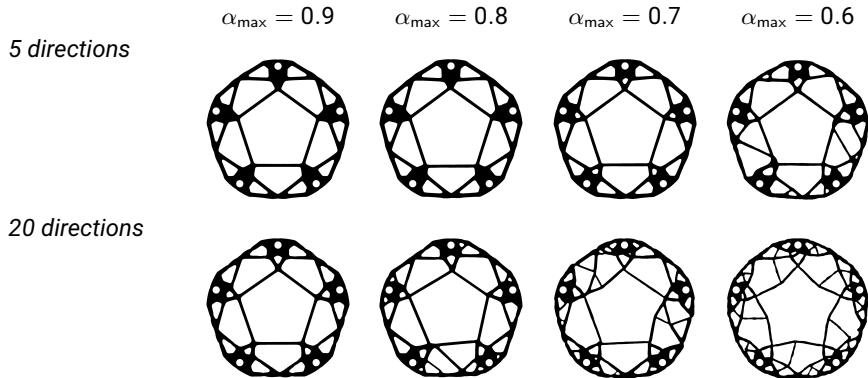
20 directions

518,144 elements

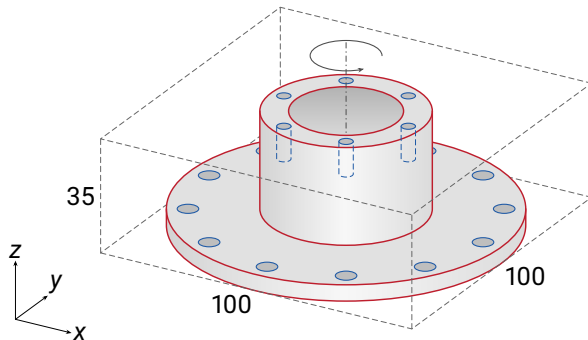
4,663,296 DOFs

$V_f = 0.4$

Vary the directions and thickness bound



Extend the inspectability constraint to 3D

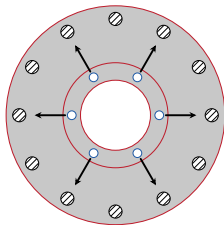


Three-dimensional bolt-circle design

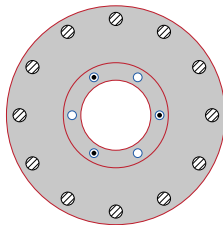
- Circular plate with a central hollow cylinder.
- Twelve perimeter sleeves are clamped.
- Six top sleeves carry the loads.

Four load cases share the same fixed supports

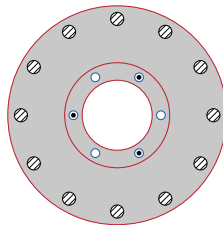
LC1: radial



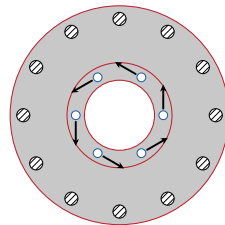
LC2: axial



LC3: axial



LC4: torsion



Hatched sleeves: $u = 0$

Dots: $+z$ loads on alternating sets of three

Inspection directions change the 3D design



Unconstrained

No thickness bound



Vertical rays

$$\alpha_{\max} = 15$$



In-plane rays

$$\alpha_{\max} = 40$$

10,682,112 hexahedral elements

32,493,375 DOFs

$$V_f = 0.3$$



Thank you

